Learning to identify new dynamical systems from a few experiments

Technical report on the tutorial notebook: pretraining, few-shot adaptation, experiment design, mismatch detection and model-based control

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Abstract

This report documents the design, mathematics, implementation and results of the notebook `few_shot_system_identification_tutorial.ipynb`. The notebook studies a family of nonlinear, partially observed, disturbed oscillators and asks whether knowledge extracted from many previously identified systems can identify a new member of the family from a few seconds of data. A two-stage pretraining procedure (per-system output-error identification followed by a principal-component model of the coefficients) yields a two-dimensional task context with an empirical Gaussian prior. New systems are identified by maximum-a-posteriori estimation of this context, with a Laplace approximation of its uncertainty. The pretrained model is then used to design informative experiments via Bayesian D-optimality, to detect systems that the learned family cannot represent, and to compute feedforward controls. All results are Monte Carlo evaluations over freshly drawn systems, and every number in this report is produced automatically from the executed notebook.

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1 Motivation and scope

Foundation models for dynamical systems aim to learn, from data of many heterogeneous systems, representations that transfer: a new system should be predicted, understood and controlled after seeing little data of its own. Three sub-problems recur in this programme:

1. **Representation.** What do systems of a kind have in common, and which few quantities distinguish one member from another?
2. **Adaptation.** How are those quantities inferred for a new system from short, noisy, partial observations, and with what uncertainty?
3. **Active data collection and reliability.** Which experiment makes the adaptation most informative, and how do we notice when the new system lies outside what was learned?

The notebook isolates these questions in a setting small enough that every component can be written in a few lines, inspected and validated against ground truth. The physics-based model class used by the learner is deliberately transparent; §12 discusses how each component maps to its learned, large-scale counterpart.

Design principles. (i) The simulator and the learner are strictly separated: the learner never accesses the true coefficients, the family structure, the velocity or the disturbance. (ii) The learner's model is *misspecified* (§2), so that no method can reach zero error. (iii) Every comparison is made over many freshly drawn systems with paired designs, and the random baselines are never selected by the criterion they are compared against.

2 The data-generating process

2.1 Continuous-time plant

Each system is a cart on a hardening spring with viscous damping, driven through a saturating actuator and perturbed by an unknown force. With position q and velocity v,

$$\dot{q} = v, \qquad \dot{v} = -kq - dv + b\,\varphi(u) - \alpha q^3 \; - \; \gamma q^5 + r + w, \qquad \varphi(u) = 0.3\tanh(u/0.6). \tag{1}$$

The coefficients $\eta = [k,\ d,\ b,\ \alpha]^\top$ are those the learner models; the quintic stiffness γ, the constant force bias r and the disturbance w are *not* part of the learner's model. The actuator limits the force to $|\varphi| < 0.3$; commands beyond $|u| \approx 1$ produce little additional force, which bounds the admissible excitation.

2.2 Discretisation

With sampling time $\Delta t = 0.1\,\mathrm{s}$ the simulator uses the semi-implicit (symplectic) Euler scheme with a one-sample input delay,

$$v_{t+1} = v_t + \Delta t\, a_t(q_t, v_t, u_{t-1}, w_t), \qquad q_{t+1} = q_t + \Delta t\, v_{t+1}, \tag{2}$$

where a_t is the right-hand side of (1). The natural frequencies in the family are $\omega_n \approx \sqrt{k} \in [0.8, 1.4]\,\mathrm{rad/s}$, so $\omega_n \Delta t \le 0.15$, well inside the stability region ($\omega \Delta t < 2$) of the scheme. Every experiment starts at rest from $q_0 = 0.2$, $v_0 = 0$; the initial state is known.

2.3 Family structure

The coefficients of a system are generated from a hidden two-dimensional design variable $c \sim \mathcal{U}([-1,1]^2)$ and an individual deviation,

$$\eta = \eta_0 + Mc + \epsilon, \quad \eta_0 = \begin{bmatrix} 1.20 \\ 0.32 \\ 0.90 \\ 1.00 \end{bmatrix}, \quad M = \begin{bmatrix} 0.55 & 0.12 \\ 0.07 & 0.11 \\ 0.16 & -0.09 \\ 0.30 & 0.45 \end{bmatrix}, \quad \epsilon \sim \mathcal{N}(0, \mathrm{diag}(0.015, 0.004, 0.008, 0.04)^2). \tag{3}$$

The deviation ϵ is *off-manifold*: it cannot be represented by any two-dimensional context. The unmodelled terms are drawn per system as $\gamma \sim \mathcal{U}[0.035, 0.047]$ and $r = 0.006\, c_1 + 0.002\, \xi$, $\xi \sim \mathcal{N}(0,1)$.

2.4 Noise and observation

The disturbance is a coloured AR(1) process and only position is measured:

$$w_t = \rho\, w_{t-1} + \sigma_w e_t, \qquad y_t = q_t + \nu_t, \qquad e_t \sim \mathcal{N}(0,1),\ \nu_t \sim \mathcal{N}(0, \sigma_y^2), \tag{4}$$

with $\rho = 0.8$, $\sigma_w = 0.01$ (stationary standard deviation $\sigma_w/\sqrt{1-\rho^2} \approx 0.017$) and $\sigma_y = 0.01$. Figure 1 shows five members of the family and one measured experiment.

2.5 Inputs

All inputs, random or designed, are drawn from the same admissible class: normalised multisines

$$u_t = \frac{s_t}{\max_\tau |s_\tau|}, \qquad s_t = \sum_{i=1}^{n} \sin(2\pi f_i t \Delta t + \phi_i), \qquad \log f_i \sim \mathcal{U}[\log 0.02,\ \log 0.8],\ \phi_i \sim \mathcal{U}[0, 2\pi], \tag{5}$$

with $n = 3$ components for experiments on new systems and $n = 5$–6 for long recordings and validation. The log-uniform frequency law covers the resonance band ($\approx$0.13–0.22 Hz) as well as frequencies far above and below it.

3 The learner's model and single-system identification

3.1 Nominal model

The learner uses (2) without γ, r and w, simulated in free run from the known initial state: $\hat{q}_t(\eta; u)$ denotes the model's position at time t under input u. The implementation (`predict`) is vectorised over arbitrary batch dimensions of η and u, which later allows all candidate experiments, contexts and finite-difference perturbations to be simulated in a single time loop.

3.2 Output-error estimation

Given one experiment of length T,

$$\hat{\eta} = \underset{\eta \in [\eta_{\min}, \eta_{\max}]}{\arg\min} \sum_{t=0}^{T} \big(y_t - \hat{q}_t(\eta; u)\big)^2, \tag{6}$$

solved with a trust-region reflective method (`scipy.optimize.least_squares`) with Jacobian scaling, box constraints $\eta_{\min} = [0.05, 0.01, 0.05, 0]$, $\eta_{\max} = [5, 2, 3, 5]$ and the generic initial guess $[1, 0.5, 1, 0.5]$. The Gauss–Newton covariance $\hat{\sigma}_i^2 (J^\top J)^{-1}$, with J the output Jacobian at the solution and $\hat{\sigma}_i^2$ the mean squared residual, is returned as a measure of estimation uncertainty. The bounds and the initial guess are deliberately *not* taken from the family (3), so that from-scratch identification uses no information about it.

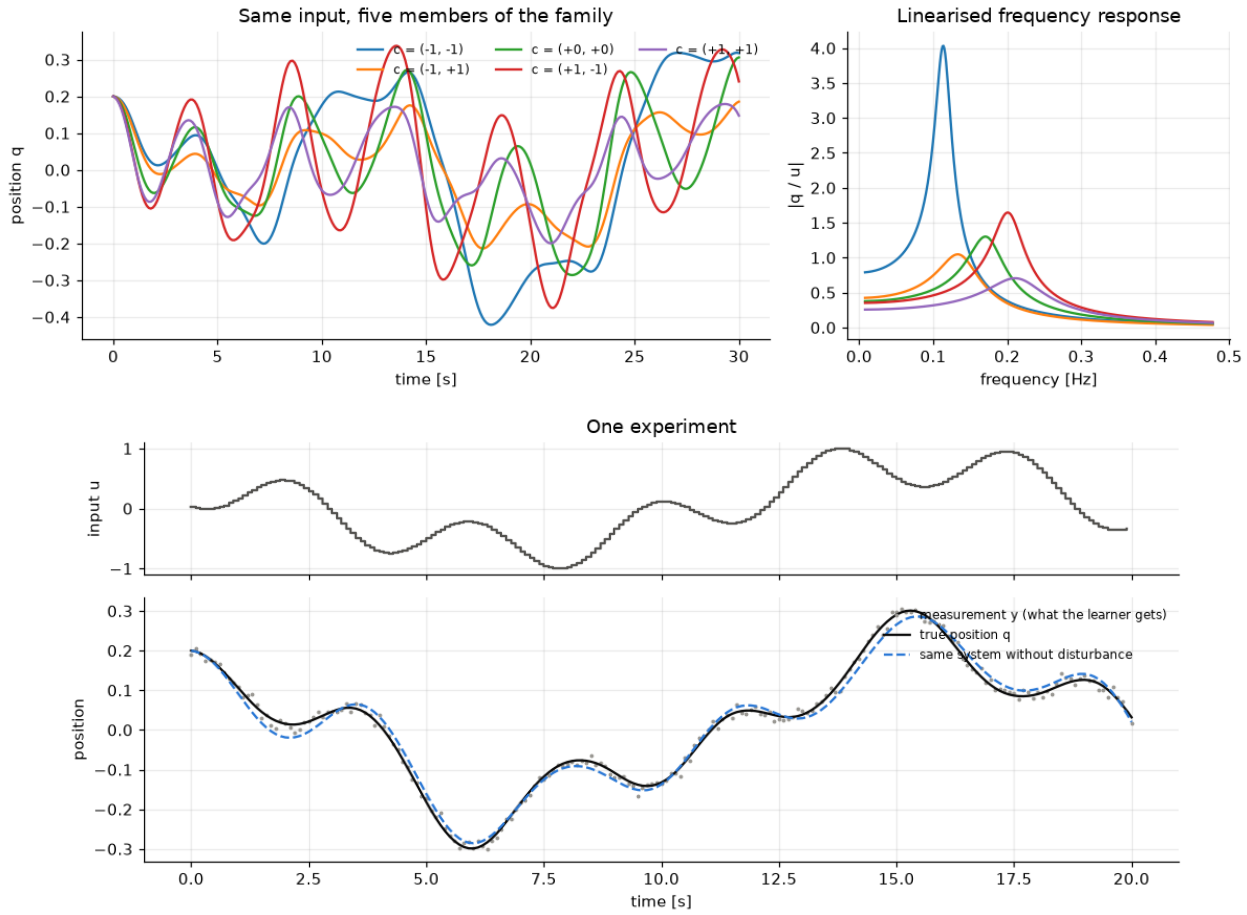


Figure 1: Top: five members of the family driven by the same input (left) and their linearised frequency responses $|0.5\,b/(k-\omega^2+\mathrm{j}d\omega)|$ (right; $0.5=\varphi'(0)$). **Bottom:** one experiment. The learner receives only the grey measurements; the gap between the true position and the disturbance-free response is caused by the coloured disturbance.

Why output error. An equation-error (regression) formulation would require the velocity and acceleration, i.e. first and second differences of y. Their noise variances scale as $\sigma_y^2/\Delta t^2$ and $\sigma_y^2/\Delta t^4$, and the regressors become noisy (errors-in-variables bias). Output error avoids differentiation and is the maximum-likelihood estimator under white measurement noise and a correct model. It is *not* maximum likelihood in the presence of process noise, for which a prediction-error method with a state estimator (e.g. an extended Kalman filter) would be appropriate. The simpler estimator is kept for transparency, and its bias is part of what the experiments measure.

3.3 Identifiability with short data

Figure 2 repeats (6) 30 times on one system with independent inputs and noise, for 4 s and 40 s of data. Table 1 reports the median relative errors.

Table 1: Median relative error of from-scratch estimates over 30 repeated experiments on one system.

Data	k	d	b	α
4 s	10.4 %	37.3 %	9.8 %	100.0 %
40 s	6.4 %	6.2 %	2.7 %	46.9 %

Four seconds are shorter than one oscillation period ($2\pi/\omega_n \approx 4.5$–$8$ s): the output sensitivities $\partial\hat{q}/\partial\eta_j$ are nearly collinear, the Gauss–Newton matrix is ill-conditioned, and many coefficient vectors explain the data equally well. The cubic coefficient α is weakly identifiable even with 40 s of data:

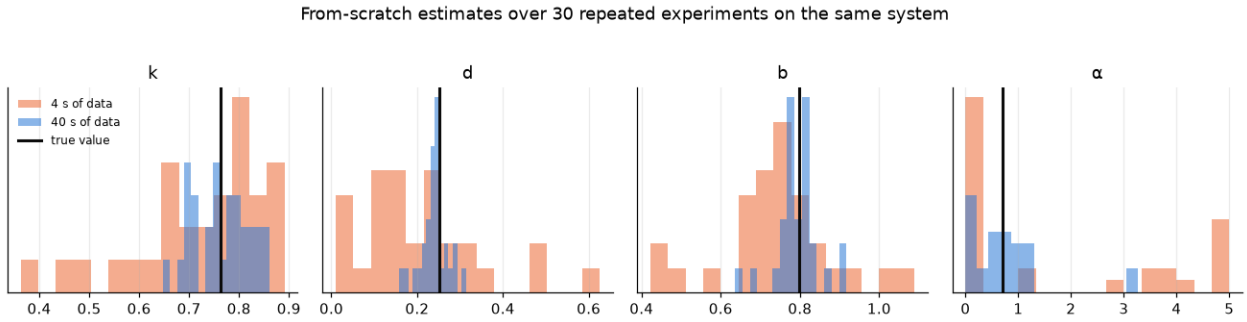


Figure 2: From-scratch estimates over 30 repeated experiments on the same system (vertical line: true value). Short experiments leave the coefficients poorly determined; α remains uncertain with long data.

with 4 s its estimates collapse onto the bounds, and with 40 s they are still off by tens of percent. Its regressor q^3 matters only at the largest deflections, and it is confounded with the unmodelled quintic term γq^5 and with the disturbance.

4 Pretraining: a low-dimensional model of the family

4.1 Two-stage procedure

Twenty training systems are drawn from the family and each is recorded for 80 s under a six-component multisine. Pretraining proceeds in two stages.

1. **Per-system identification:** $\hat{\eta}_i$ and its Gauss–Newton covariance P_i from (6), $i = 1, \dots, N$, $N = 20$.
2. **Noise-whitened principal component model.** Let $s_j = \left(\frac{1}{N}\sum_i (P_i)_{jj}\right)^{1/2}$ be the average estimation standard deviation of coefficient j and $D = \mathrm{diag}(s)$. With $\bar{\eta} = \frac{1}{N}\sum_i \hat{\eta}_i$ and the SVD of the scaled, centred matrix $X = [D^{-1}(\hat{\eta}_1 - \bar{\eta}), \dots, D^{-1}(\hat{\eta}_N - \bar{\eta})]^\top = W\Sigma V^\top$, the sample covariance of X has eigenvalues $\lambda_j = \Sigma_{jj}^2/(N-1)$. For rank $r = 2$,

$$U = D\,V_{:,1:r}\,\Lambda_r^{1/2}, \qquad z_i = \Lambda_r^{-1/2} V_{:,1:r}^\top D^{-1}(\hat{\eta}_i - \bar{\eta}), \qquad \hat{\eta}_i \approx \bar{\eta} + U z_i. \tag{7}$$

Why noise whitening. The spread of the estimates $\hat{\eta}_i$ has two sources: genuine differences between systems and estimation error. Plain PCA ($D = I$) ranks directions by raw spread and therefore allocates a principal direction to the scatter of the poorly identified α, whose estimation standard deviation is one to two orders of magnitude larger than those of k, d and b (here $s = [0.0111, 0.0054, 0.0069, 0.3616]$). Scaling by s ranks directions by signal-to-noise ratio instead. This is the homoscedastic special case of factor analysis with known noise variances, and a first-order substitute for the joint hierarchical model discussed below.

The whitening by $\Lambda_r^{-1/2}$ makes the training contexts have identity sample covariance, so $z \sim \mathcal{N}(0, I)$ is the Gaussian population model implied by the training set: an empirical-Bayes prior for new systems. Finally, the pooled residual level $\hat{\sigma}^2 = \frac{1}{N}\sum_i \frac{1}{T+1}\sum_t (y_{i,t} - \hat{q}_t(\hat{\eta}_i))^2$ estimates the effective noise the adapted model should expect. It includes measurement noise, the effect of the disturbance on the free-run output, and model mismatch, and is therefore larger than σ_y.

4.2 Results

The first two principal directions explain 78.4 % and 19.4 % of the noise-weighted variance (97.8 % together), and $\hat{\sigma} = 0.0184$ (versus $\sigma_y = 0.01$). Figure 3 shows that k, d and b are recovered well from long recordings while α is only roughly determined, and that the learned contexts vary smoothly with the hidden design variables. They are a rotated and rescaled version of c, which is all that can be expected: the likelihood is invariant to invertible reparameterisations of the context.

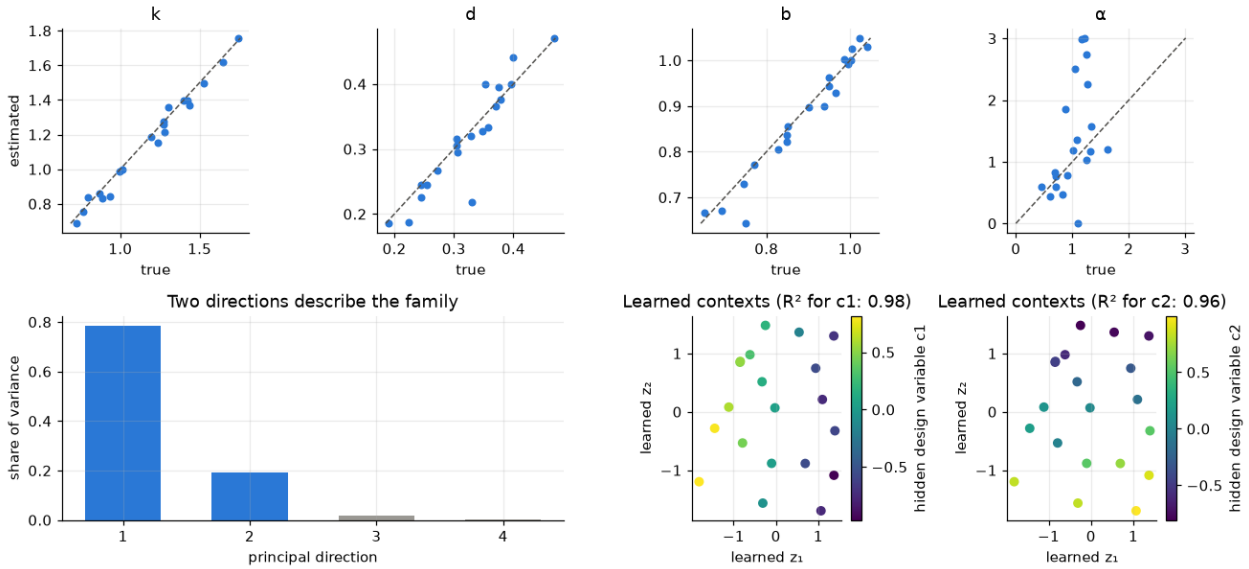


Figure 3: Pretraining. **Top:** per-system estimates versus truth. **Bottom left:** explained variance per principal direction. **Bottom right:** learned contexts coloured by the hidden design variables c_1, c_2.

Table 2 quantifies how much of each hidden design variable is explained by an affine function of the learned contexts. Noise whitening is decisive for the second design variable, which acts mainly through the poorly identified α and the small damping variations.

Table 2: Coefficient of determination R^2 of the hidden design variables regressed (affinely) on the learned training contexts.

Family model	c_1	c_2
plain PCA	0.96	0.35
noise-whitened PCA	0.98	0.96

Relation to hierarchical models. The two-stage estimator approximates a linear mixed-effects (hierarchical Bayesian) model $\eta_i = \bar{\eta} + U z_i + \epsilon_i$, $z_i \sim \mathcal{N}(0, I)$, fitted by treating the stage-one estimates as data. A joint estimator (e.g. expectation–maximisation over all trajectories) would propagate the full per-system estimation covariances P_i, including their correlations and their variation across systems. Noise-whitened PCA uses only the average marginal variances, which suffices here to stop poorly identified coefficients from inflating the apparent family variance.

5 Few-shot adaptation

5.1 MAP estimation of the context

For a new system, $\bar{\eta}$, U and $\hat{\sigma}$ are frozen and only z is estimated:

$$\hat{z} = \arg\min_{z \in \mathbb{R}^2} \frac{1}{\hat{\sigma}^2} \sum_{t=0}^{T} \left(y_t - \hat{q}_t(\mathcal{D}(z); u)\right)^2 + \|z\|^2, \qquad \mathcal{D}(z) = \mathrm{clip}\left(\bar{\eta} + Uz,\ \eta_{\min}, \eta_{\max}\right). \tag{8}$$

The problem is solved as a stacked least-squares problem with residual $[\,(y - \hat{q}(\mathcal{D}(z)))/\hat{\sigma};\ z\,]$ from $z = 0$ (the population mean). The clipping keeps decoded coefficients physical, e.g. $\alpha \geq 0$; a negative hardening coefficient would make the model diverge for large deflections.

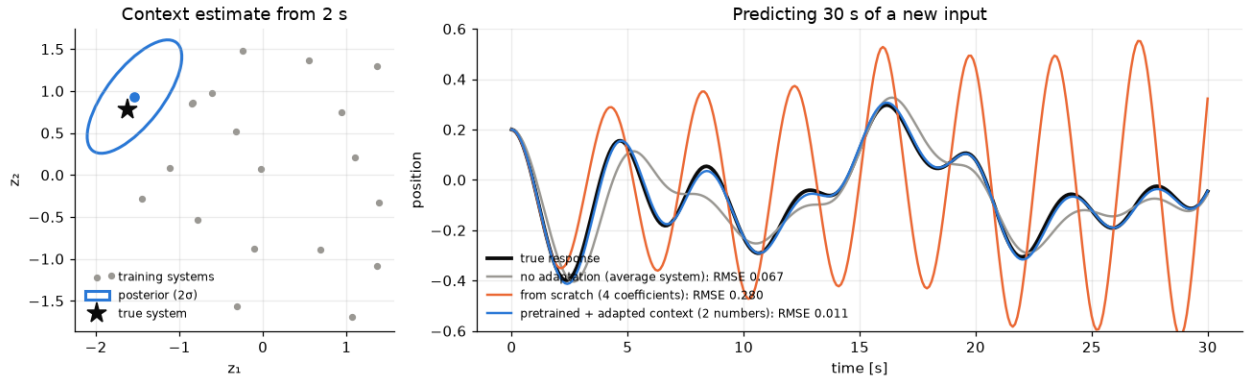


Figure 4: Adaptation from 2 s of data. **Left:** MAP context, 2σ Laplace ellipse, and the projection of the true coefficients onto the learned plane (star). **Right:** prediction of the disturbance-free response to a new 30 s input.

5.2 Laplace approximation

With $S = \partial\hat{q}/\partial z \in \mathbb{R}^{(T+1)\times 2}$ evaluated at $\hat{z}$ (finite differences inside the solver), the Gauss–Newton approximation of the negative log-posterior Hessian gives

$$\mathrm{Cov}(z \mid y) \approx \left(\hat{\sigma}^{-2} S^\top S + I\right)^{-1}. \tag{9}$$

This expression assumes independent residuals. With coloured disturbances the effective number of independent samples is smaller than $T+1$, so (9) is expected to be overconfident (§11).

5.3 Why it helps

Two effects combine. The number of unknowns drops from four to two, which lowers variance. The prior restricts estimates to the region occupied by the family, which regularises directions the data cannot resolve. The price is bias whenever the new system is not representable as $\bar{\eta} + Uz$: the off-manifold deviation ϵ, the stage-one estimation errors, and in §9 deliberate shifts.

In the single example of Figure 4 (2 s of data, a third of an oscillation period), the validation RMSE is 0.0668 with no adaptation, 0.2799 from scratch and 0.0109 with the adapted context. The from-scratch fit explains the short record with the coefficients of a much less damped oscillator, and its free-run prediction drifts away. The adapted context stays within the family and extrapolates well.

6 Evaluation protocol

- **Fresh systems.** Every trial draws a new system from (3); no test system is used for pretraining.
- **Metric.** Models are compared on a fresh 30 s, six-component multisine by the RMSE between the model prediction $\hat{q}_t(\hat{\eta})$ and the system's *disturbance-free* response ($w \equiv 0$, $\nu \equiv 0$, but γ and r present). This isolates model quality from the unpredictable disturbance realisation.
- **Reference.** The true η inserted in the nominal model gives the error due to γ and r alone. It is a reference, not a lower bound: a fitted model can partly compensate unmodelled terms.
- **Pairing.** Within a trial all methods are evaluated on the same system and the same validation input. In the data-budget study they also share the same experiment; in the design study each method runs its own experiment with independent noise.
- **Fair baselines.** Random experiments are drawn from the same admissible class and are never scored. Designed experiments are selected before the new system is excited.
- **Uncertainty.** Medians with inter-quartile ranges, paired win rates, and percentile bootstrap intervals (5000 resamples) of mean paired differences.

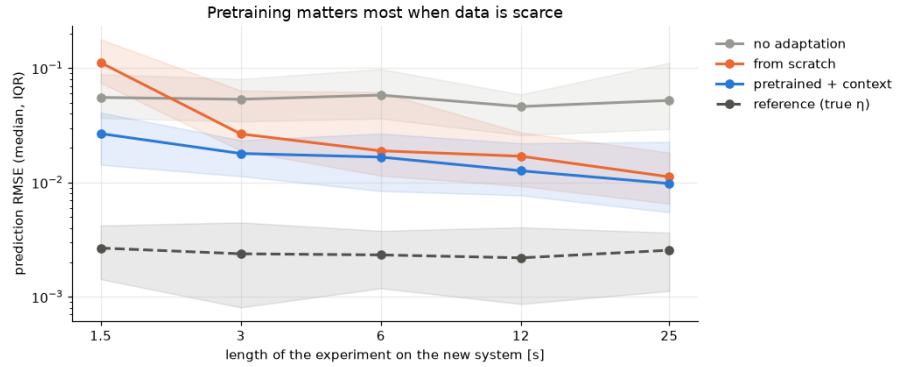


Figure 5: Validation error against the length of the experiment on the new system (log–log; lines: medians, bands: inter-quartile ranges).

7 Data-budget study

For $T \in \{15, 30, 60, 120, 250\}$ samples (1.5–25 s) and 40 systems per budget, one random three-component experiment is run per system and four predictors are compared: no adaptation ($\bar{\eta}$), from scratch (6), pretrained context (8), and the reference.

Table 3: Validation RMSE, median [IQR] over 40 new systems per budget. Last column: ratio of the from-scratch to the pretrained median.

$T\Delta t$ [s]	no adaptation	from scratch	pretrained + context	reference	ratio
1.5	0.0554 [0.037, 0.089]	0.1118 [0.074, 0.179]	0.0267 [0.014, 0.041]	0.0027 [0.001, 0.004]	4.2
3	0.0534 [0.034, 0.080]	0.0266 [0.019, 0.064]	0.0179 [0.011, 0.023]	0.0024 [0.001, 0.004]	1.5
6	0.0581 [0.036, 0.098]	0.0189 [0.011, 0.062]	0.0167 [0.008, 0.027]	0.0023 [0.001, 0.004]	1.1
12	0.0461 [0.026, 0.059]	0.0170 [0.009, 0.027]	0.0126 [0.008, 0.022]	0.0022 [0.001, 0.004]	1.3
25	0.0522 [0.029, 0.111]	0.0112 [0.007, 0.018]	0.0098 [0.005, 0.023]	0.0026 [0.001, 0.004]	1.1

Table 3 and Figure 5 show the characteristic few-shot pattern. With scarce data, from-scratch identification overfits and is worse than using the population mean, while the pretrained context already halves the error of the population mean. As the budget grows, the from-scratch estimator improves faster and the ratio in the last column falls from about four towards one: pretraining buys *data efficiency*, and in this setting it remains at least as good as identification from scratch at every budget tested. Both learned predictors remain above the reference. The residual gap is structural (stage-one errors in $\bar{\eta}, U$, the rank restriction, and the poorly determined α) and cannot be removed by more data on the new system.

8 Experiment design

8.1 Criterion

For additive Gaussian output noise, the Fisher information of the context under input u, including the unit prior precision, is

$$\mathcal{I}(u; z) = \hat{\sigma}^{-2} S(u; z)^\top S(u; z) + I, \qquad S_{tj}(u; z) = \frac{\partial \hat{q}_t(\mathcal{D}(z); u)}{\partial z_j}, \tag{10}$$

the inverse of the Laplace covariance (9) evaluated before the data are seen. Since z is unknown, the design maximises the prior expectation of the log-determinant (Bayesian D-optimality), approximating the prior by the $N = 20$ training contexts:

$$u^\star = \arg\max_{u \in \mathcal{C}} \; J(u), \qquad J(u) = \frac{1}{N} \sum_{i=1}^{N} \log\det \mathcal{I}(u; z_i). \tag{11}$$

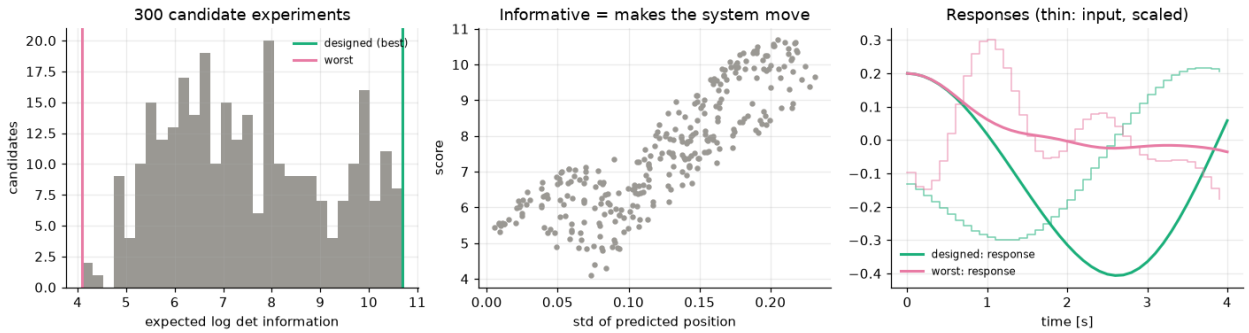


Figure 6: Left: scores of 300 candidate experiments. **Centre:** score versus the standard deviation of the predicted motion of the average system. **Right:** responses of the average system to the best and worst candidate (thin lines: scaled inputs).

For a model linearised around z, $\frac{1}{2}\log\det\mathcal{I}(u;z)$ equals the expected information gain (mutual information between z and y) under the prior $\mathcal{N}(0,I)$, so J is a local-linear approximation of the expected information gain of the experiment.

8.2 Implementation

The sensitivities are computed by central differences ($\varepsilon = 10^{-4}$) in both context directions. One vectorised call of the model simulates all $|\mathcal{C}| \times N \times 2 \times 2$ trajectories in a single loop over time, so scoring 200–300 candidates takes a fraction of a second. The optimiser is random search: $u^\star$ is the best of a pool $\mathcal{C}$ of admissible multisines (5). For illustration, the worst candidate ($\arg\min J$) is also retained.

8.3 Physical interpretation

Figure 6 shows the score distribution of 300 four-second candidates. The score correlates with how much the candidate moves the average system (correlation 0.85), since a system that does not move produces outputs that are insensitive to its coefficients. The best candidates swing the cart through large, two-sided deflections at low frequency and near resonance. The worst shake it at high frequency, which the mass filters out. The correlation is not perfect: the *shape* of the motion determines which directions of z become distinguishable. In the example of Figure 7, the determinant of the posterior covariance is 5.0×10^{-3} (worst), 2.6×10^{-4} (random) and 1.1×10^{-5} (designed).

The same figure exposes a limitation: the smallest (designed) ellipse does not contain the projection of the true coefficients. The Laplace posterior is overconfident, because (9) treats the residuals as white and the model as correct, while the disturbance is coloured and γ, r are missing. A smaller ellipse therefore signals more information, but its absolute size cannot be trusted as a calibrated uncertainty (§11).

8.4 Monte Carlo evaluation

For $T \in \{15, 30, 60, 120\}$ and 60 new systems per budget, each trial draws a fresh pool of 200 candidates, selects the designed and the worst experiment, draws an unscored random experiment, and adapts the context from each. Table 4 reports the mean validation RMSE, the relative reduction of the designed over the random experiment, the paired win rate, and a bootstrap interval for the mean paired reduction.

The value of design depends on the budget. With 1.5 s (a quarter of an oscillation period) no experiment reveals much, the prior dominates, and the difference between designed and random experiments is not significant: the bootstrap interval contains zero. From 3 s on, the designed experiment lowers the mean error by roughly a third to a half, wins about three quarters of the paired trials, and all bootstrap intervals exclude zero, while the worst experiments waste most of the

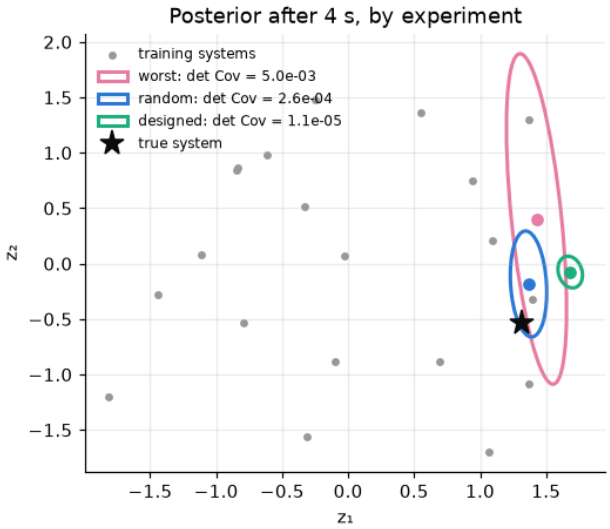


Figure 7: Posterior (2σ Laplace ellipses) for one new system after 4 s under the worst, a random and the designed experiment (star: projection of the true coefficients).

Table 4: Pretrained context adaptation under different experiments (60 new systems per budget).

$T\Delta t$ [s]	worst	random	designed	reduction	designed wins	95 % CI (random − designed)
1.5	0.0267	0.0279	0.0317	-13.6 %	48 %	[−0.0096, +0.0013]
3	0.0301	0.0244	0.0126	+48.3 %	73 %	[+0.0076, +0.0164]
6	0.0414	0.0213	0.0126	+40.8 %	78 %	[+0.0059, +0.0118]
12	0.0531	0.0160	0.0111	+30.8 %	73 %	[+0.0018, +0.0086]

data. The relative advantage is largest for the shortest informative experiments and decreases as the experiment grows, since random inputs then also accumulate enough information. The designed experiment does not win every paired trial. The criterion targets parameter uncertainty under a linearised, white-noise model rather than the validation error itself, and mismatch and coloured disturbances remain.

9 Detecting systems outside the learned family

9.1 Construction of out-of-family systems

Let $V_\perp \in \mathbb{R}^{4\times 2}$ be an orthonormal basis of the orthogonal complement of span(U) (from a full SVD of U). The column with the largest damping component, signed to increase damping, defines a direction n that no context can express. The direction realised in the run is $n = [-0.16, 0.97, -0.13, -0.15]^\top$ in $[k, d, b, \alpha]$. Test systems are drawn from the family and shifted, $\eta \leftarrow \text{clip}(\eta + \delta n)$, for seven values $\delta \in \{0, 0.1, \ldots, 0.6\}$, 30 systems each, with 12 s random experiments. The shift is measured in raw coefficient units, which weights coefficients by their natural scale; a normalised metric would be a natural refinement.

9.2 Misfit test

After adaptation, the RMS output misfit $m = \left(\frac{1}{T+1}\sum_t (y_t - \hat{q}_t(\mathcal{D}(\hat{z})))^2\right)^{1/2}$ is compared with the threshold $\tau = 0.0215$, the empirical 95 % quantile of m over 60 in-family systems under the same protocol (nominal false-alarm rate 5 %; realised rate on the calibration set: 5 %). A system is flagged if $m > \tau$.

Table 5 and Figure 9 show the trade-off. The pretrained model is only as good as its family: as δ grows, its error grows because the prior and the two-dimensional parameterisation force a biased

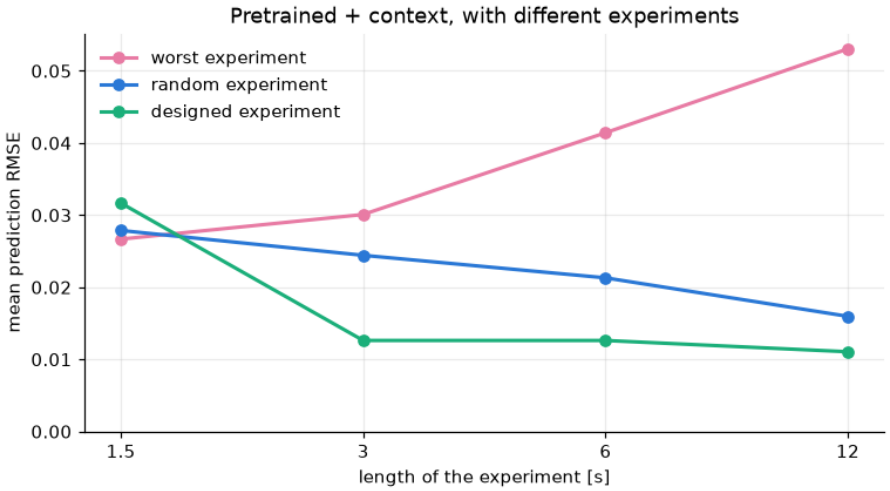


Figure 8: Mean validation RMSE of the adapted model for the worst, random and designed experiment.

Table 5: Out-of-family shifts: median validation RMSE and share of systems flagged by the misfit test.

δ	from scratch	pretrained + context	flagged
0.00	0.0164	0.0145	3 %
0.10	0.0157	0.0175	23 %
0.20	0.0093	0.0198	33 %
0.30	0.0078	0.0259	50 %
0.40	0.0078	0.0257	73 %
0.50	0.0074	0.0295	73 %
0.60	0.0069	0.0341	67 %

estimate. The from-scratch estimator makes no family assumption and is insensitive to δ. Its error even decreases, as more strongly damped systems move less. With 12 s of data from scratch is already competitive inside the family, so the pretrained model loses its advantage after a small shift. The misfit statistic grows with δ because the best representable model can no longer explain the data. The flag rate rises from the nominal false-alarm level at $\delta = 0$ to roughly three quarters of the systems for large shifts. Detection, however, lags degradation: at moderate shifts the adapter is already worse than from scratch while most systems still pass the test, because the disturbance-induced misfit varies strongly from system to system. The resulting switching rule between the small adapter and full identification is sound in principle, but its power is limited; innovation-based tests that whiten the coloured residuals would be a natural improvement.

10 Model-based feedforward control

To test whether model quality carries over to control, each candidate model computes a feedforward input that tracks a smooth reference r_t (logistic transitions $0.2 \to -0.15 \to 0.1$ over 12 s):

$$u^\star(\hat{\eta}) = \underset{u \in [-2,2]^T}{\arg\min} \sum_{t=0}^{T} \left(\hat{q}_t(\hat{\eta}; u) - r_t\right)^2 + \lambda \sum_{t=0}^{T-2} (u_{t+1} - u_t)^2, \qquad \lambda = 0.05. \tag{12}$$

The $T = 120$ decision variables are optimised by trust-region least squares with a Jacobian obtained by forward differences, computed in one vectorised batch of $T + 1$ model rollouts. The input is then applied open loop to the *true* system, including disturbances. Open-loop feedforward cannot correct model errors during execution, which makes the tracking error a direct measure of model quality. The adapted model uses one designed 6 s experiment per system.

Over 15 new systems the tracking RMSE (mean $\pm$ standard deviation) is 0.0383 ± 0.0171 with the average-system model, 0.0163 ± 0.0053 with the adapted model, and 0.0170 ± 0.0046 with the

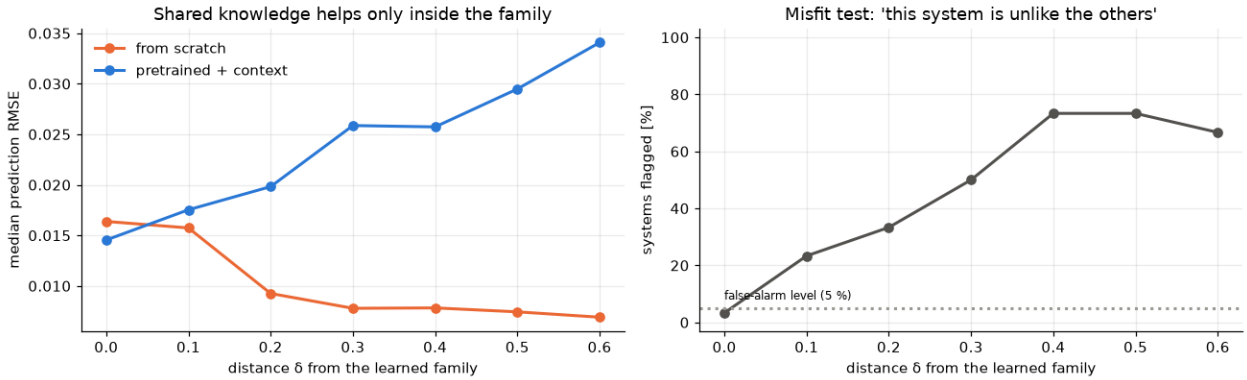


Figure 9: Left: prediction error as the new system moves away from the learned family. **Right:** share of systems flagged by the misfit test (dotted: nominal false-alarm level).

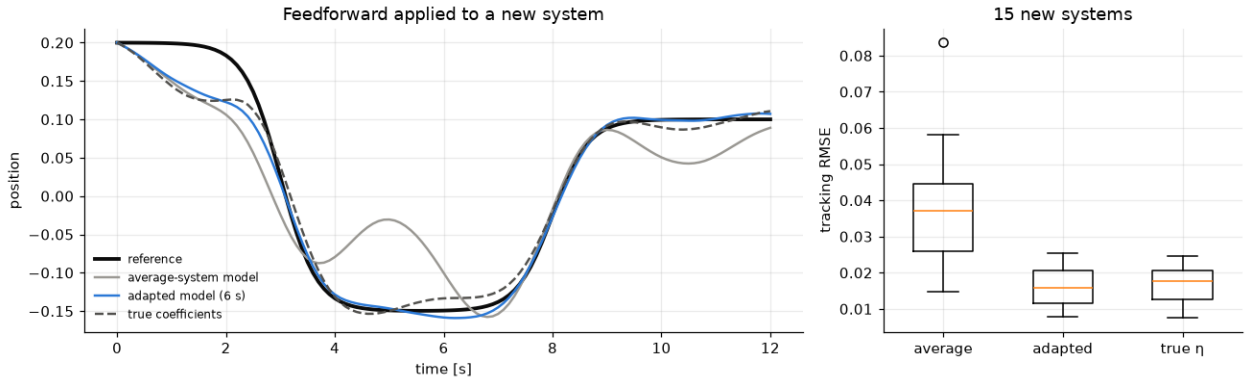


Figure 10: Left: feedforward tracking on one new system with three models. **Right:** tracking RMSE over 15 new systems.

true coefficients (Figure 10). Six seconds of designed data more than halve the tracking error of the average-system model and match the true coefficients. The adapted model can be marginally better, since it is fitted to the actual behaviour and partly compensates the unmodelled terms. The remaining error stems from the disturbance and the unmodelled terms, which only feedback can reject.

11 Limitations

- **Known model class.** The learner knows the structure of the dominant physics. A learned backbone (neural state-space or sequence model) removes this assumption but makes identifiability and extrapolation harder to analyse.
- **Two-stage pretraining.** Stage-one errors (notably in α) enter $\bar{\eta}$ and U as if they were genuine family variation. Noise-whitened PCA mitigates this with average marginal variances only; a joint hierarchical estimator would use the full per-system covariances.
- **Output error under process noise.** The estimator is not maximum likelihood when disturbances act on the state. Prediction-error methods with state estimation are the principled alternative.
- **Calibration.** The Laplace covariance assumes white residuals, while the disturbance is coloured and the model misspecified; Figure 7 shows an ellipse that misses the true system. Coverage should be measured, not assumed.
- **Design.** The criterion is local (linearised), parameter-oriented (D-optimal) and optimised by random search over one input class. It does not account for safety constraints on state or input

beyond the actuator limit.

- **Linear family model.** PCA represents the family as an affine plane; curved families require nonlinear decoders (e.g. variational autoencoders or hypernetworks).
- **Scale.** One family, one seed and moderate Monte Carlo sizes; conclusions are illustrative, not general.

12 From the tutorial to foundation models

Component	In the notebook	Scalable counterpart
Model class	known physics, 4 coefficients	structured neural state-space / sequence models
Pretraining	per-system identification + noise-whitened PCA	joint training across heterogeneous systems and sensors
Task context	$z \in \mathbb{R}^2$, prior $\mathcal{N}(0, I)$	learned context encoders, low-rank adapters, hypernetworks
Adaptation	MAP + Laplace	amortised inference, fast gradient adaptation, in-context learning
Experiment design	Bayesian D-optimal random search	safe, constrained, sequential Bayesian design
Reliability	misfit test with calibrated threshold	calibrated uncertainty, out-of-distribution detection
Use	open-loop feedforward	MPC, reinforcement learning, safe learning-based control

Research questions raised by the experiments.

1. *Identifiability of learned contexts:* under which conditions on the backbone and the excitation can the context of a new system be recovered from partial, noisy observations, and with what sample complexity?
2. *Calibrated uncertainty under coloured noise and mismatch:* how can posteriors over contexts be made trustworthy enough for decisions (design, control, safety)?
3. *Safe and goal-oriented excitation:* how should information gain be traded against state/input constraints and against the downstream task (prediction, control)?
4. *Adaptive capacity:* how can a model decide online that its adapter is insufficient, and grow it with guarantees on false alarms?
5. *Closing the loop:* how should identification, uncertainty and control be designed jointly, e.g. dual control, with stability guarantees for the adapted model?

A Reproducibility

- **Build.** `python tools/build_report.py` regenerates the notebook from `tools/build_tutorial_notebook.py`, executes it, exports the figures and `report/tutorial_metrics.json`, writes `report/results.tex` and compiles this document. All numbers in the text and tables are macros from that file.
- **Randomness.** A single generator seeded with 2026 drives all draws, in notebook order.
- **Dependencies.** NumPy, SciPy, Matplotlib; for building: nbformat, nbclient, ipykernel, pdflatex.
- **Runtime.** About two minutes on a laptop CPU, dominated by the Monte Carlo studies.

Table 6: Default configuration.

Sampling time	$\Delta t = 0.1\,\mathrm{s}$, initial state $(0.2, 0)$
Noise	$\sigma_y = 0.01$, $\sigma_w = 0.01$, $\rho = 0.8$
Pretraining	20 systems $\times$ 40 s, six-component multisines, rank 2
Data-budget study	$T \in \{15, 30, 60, 120, 250\}$, 40 systems each
Design study	$T \in \{15, 30, 60, 120\}$, 60 systems each, 200 candidates per trial
Out-of-family study	7 shifts $\delta \in [0, 0.6]$, 30 systems each, $T = 120$; 60 calibration systems
Control study	15 systems, $T = 120$, $\lambda = 0.05$, 6 s designed identification experiment
Validation	fresh 30 s six-component multisine, disturbance-free response